Independent Component Analysis

1 Introduction

Independent component analysis (ICA) is a computational method for separating a multivariate signal into additive subcomponents. This is done by assuming that the subcomponents are non-Gaussian signals and that they are statistically independent from each other. ICA is a special case of blind source separation.

ICA finds the underlying components or sources from multivariate statistical data. Given a set of observations of random variables, where each observation is a linear mixture of independent source signals, ICA recovers the original source signals without knowing the mixing coefficients.

1.1 Definition

The ICA model assumes a linear mixing process:

$$\mathbf{x} = \mathbf{A}\mathbf{s} \tag{1}$$

where:

- $\mathbf{x} = (x_1, x_2, \ldots, x_m)^T$ is the observed random vector
- $\mathbf{s} = (s_1, s_2, \ldots, s_n)^T$ is the independent component vector
- $\mathbf{A}$ is the $m \times n$ mixing matrix

The goal is to estimate both $\mathbf{A}$ and $\mathbf{s}$ from $\mathbf{x}$ by finding a separating matrix $\mathbf{W}$ such that:

$$\mathbf{s} = \mathbf{W}\mathbf{x} \tag{2}$$

2 ICA Dimensions Clarification

2.1 Understanding the Vector Form

The ICA model $\mathbf{x} = \mathbf{A}\mathbf{s}$ describes what happens at **one time instant**:

$$
\underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}}_{\text{m sensors}} = \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_{\text{mixing matrix}} \underbrace{\begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}}_{\text{n sources}} \tag{3}
$$

2.2 Concrete Example: Audio Separation

- 2 microphones in a room ($m = 2$)
- 2 people speaking ($n = 2$)
- At time t:
  - $x_1(t)$: sound level at microphone 1
  - $x_2(t)$: sound level at microphone 2
  - $s_1(t)$: speaker 1's voice amplitude
  - $s_2(t)$: speaker 2's voice amplitude
- $\mathbf{A}$: describes how sound propagates from speakers to microphones

2.3 Complete Data Matrix

In practice, we have many time samples:

$$
\underbrace{\begin{bmatrix} x_1(1) & x_1(2) & \cdots & x_1(N) \\ x_2(1) & x_2(2) & \cdots & x_2(N) \\ \vdots & \vdots & \ddots & \vdots \\ x_m(1) & x_m(2) & \cdots & x_m(N) \end{bmatrix}}_{\mathbf{X} \in \mathbb{R}^{m \times N}} = \mathbf{A} \underbrace{\begin{bmatrix} s_1(1) & s_1(2) & \cdots & s_1(N) \\ s_2(1) & s_2(2) & \cdots & s_2(N) \\ \vdots & \vdots & \ddots & \vdots \\ s_n(1) & s_n(2) & \cdots & s_n(N) \end{bmatrix}}_{\mathbf{S} \in \mathbb{R}^{n \times N}} \tag{4}
$$

2.4 Key Assumptions

For ICA to be possible, the following assumptions must hold:

1. **Statistical independence**: The components s_i are statistically independent.
2. **Non-Gaussian distribution**: The independent components must have non-Gaussian distributions (with the possible exception of one Gaussian component).
3. **Mixing matrix invertibility**: The mixing matrix $\mathbf{A}$ must be square and invertible, or the number of observed mixtures m must be at least equal to the number of sources n.

3 Mathematical Foundations

3.1 Statistical Independence and Mutual Information

Two random variables y_1 and y_2 are independent if their joint probability density function (pdf) factorizes:

$$p(y_1, y_2) = p_1(y_1)p_2(y_2) \tag{5}$$

For multiple variables, statistical independence means:

$$p(y_1, y_2, \ldots, y_n) = \prod_{i=1}^{n} p_i(y_i) \tag{6}$$

3.1.1 Mutual Information

Mutual information measures the statistical dependence between random variables. For two continuous random variables X and Y, the mutual information is defined as:

$$I(X;Y) = \iint p(x,y) \log \frac{p(x,y)}{p(x)p(y)} dxdy \tag{7}$$

For multiple random variables $\mathbf{y} = (y_1, y_2, \ldots, y_n)$, the mutual information is:

$$I(\mathbf{y}) = \int p(\mathbf{y}) \log \frac{p(\mathbf{y})}{\prod_{i=1}^{n} p_i(y_i)} d\mathbf{y} \tag{8}$$

Mutual information can also be expressed in terms of entropy:

$$I(y_1, y_2, \ldots, y_n) = \sum_{i=1}^{n} H(y_i) - H(\mathbf{y}) \tag{9}$$

where $H(y_i)$ is the marginal entropy of y_i and $H(\mathbf{y})$ is the joint entropy.

In ICA, minimizing mutual information between components is equivalent to making them statistically independent, since $I(\mathbf{y}) = 0$ if and only if the components are independent.

3.2 Non-Gaussianity and the Central Limit Theorem

The central limit theorem provides the key insight: a sum of independent random variables tends toward a Gaussian distribution. Therefore, to find independent components, we look for the most non-Gaussian directions.

3.3 Measures of Non-Gaussianity

3.3.1 Kurtosis

Kurtosis is the classical measure of non-Gaussianity, defined as the fourth-order cumulant:

$$\text{kurt}(y) = E\{y^4\} - 3(E\{y^2\})^2 \tag{10}$$

For Gaussian variables, kurtosis is zero. Super-Gaussian distributions have positive kurtosis, sub-Gaussian have negative.

3.3.2 Negentropy

A more robust measure is negentropy, based on the information-theoretic quantity of entropy:

$$J(y) = H(y_{\text{gauss}}) - H(y) \tag{11}$$

where $H(y) = -\int p(y) \log p(y) dy$ is the differential entropy, and y_{gauss} is a Gaussian random variable with the same covariance matrix as y.

4 Preprocessing for ICA

4.1 Centering

The first step is to center the observed data by subtracting the mean:

$$\mathbf{x} \leftarrow \mathbf{x} - E\{\mathbf{x}\} \tag{12}$$

4.2 Whitening

Whitening transforms the data so that its components are uncorrelated and have unit variance:

$$\mathbf{z} = \mathbf{D}^{-1/2}\mathbf{E}^T\mathbf{x} \tag{13}$$

where $\mathbf{E}$ is the matrix of eigenvectors of the covariance matrix $E\{\mathbf{x}\mathbf{x}^T\}$, and $\mathbf{D}$ is the diagonal matrix of eigenvalues.

After whitening, $E\{\mathbf{z}\mathbf{z}^T\} = \mathbf{I}$.

5 ICA Algorithms

5.1 FastICA

The FastICA algorithm is one of the most popular ICA methods. It uses a fixed-point iteration scheme to find directions that maximize non-Gaussianity.

For one unit, the algorithm computes:

$$\mathbf{w}^{+} = E\{\mathbf{z}g(\mathbf{w}^T\mathbf{z})\} - E\{g'(\mathbf{w}^T\mathbf{z})\}\mathbf{w} \tag{14}$$

$$\mathbf{w} \leftarrow \mathbf{w}^{+}/\|\mathbf{w}^{+}\| \tag{15}$$

where g is a non-linear function, typically $g(u) = \tanh(u)$ or $g(u) = u^3$.

5.2 Infomax

The Infomax approach maximizes the mutual information between the inputs and outputs of a neural network, which is equivalent to minimizing the mutual information between the output components.

6 Applications

ICA has been successfully applied in numerous fields:

- **Signal processing**: Separation of audio signals, removal of artifacts from EEG and MEG data
- **Image processing**: Feature extraction, texture analysis
- **Biomedical data analysis**: Analysis of functional MRI data, genomic data
- **Econometrics**: Analysis of financial time series

7 Limitations and Considerations

- ICA cannot determine the order of independent components (permutation ambiguity)
- ICA cannot determine the variance (scale) of independent components (scaling ambiguity)
- The number of sources must be known or estimated
- All sources (except possibly one) must be non-Gaussian
- The mixing must be linear and instantaneous

8 Conclusion

Independent component analysis provides a powerful framework for blind source separation and feature extraction. By exploiting the statistical independence and non-Gaussianity of source signals, ICA can recover underlying components from observed mixtures. The connection between minimizing mutual information and achieving statistical independence provides a fundamental theoretical basis for ICA algorithms.